

Demonstrating *Commons Sense*: A Novel Physiological Interaction For Engaging Live Streaming Audiences

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ABSTRACT

Now more than ever, people are using online platforms to communicate. Twitch, the foremost platform for live game streaming, offers many communication modalities. However, the platform lacks representation of social cues and signals of the audience experience, which are innately present in live events. For this demonstration, we present an interactive experience that captures the audience energy and response in a game streaming context. We designed a survival shooter game and integrated a custom-communication modality (*Commons Sense*) in which remote audience attendees' heart rates will be sensed via webcam, averaged, and fed into a video game as input to affect sound, lighting, and difficulty.

CCS CONCEPTS

• **Human-centered computing** → *Collaborative interaction*; • **Software and its engineering** → *Interactive games*.

KEYWORDS

physiology, live streaming, Twitch, affective games, audience participation, heart rate

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1 INTRODUCTION

Interacting with others and forming relationships is essential for mental and physical well-being. However, those who love each other, know one another, or work together are not always geographically located near to one another. To help non co-located people

stay connected [6], there has been a rise in online platforms that facilitate peer-to-peer synchronous communication (e.g., WhatsApp¹, Twitch², and teleconferencing applications like Zoom³). Twitch is the leading platform for live streaming video games, with viewership rising each year at 2.9 million average viewers per day [17]. In these online platforms, many social cues and signals inherent to face-to-face (f2f) communication (e.g., para-linguistic cues, facial and body language and posture, eye contact) are compromised, including the ability of the streamer to see the audience responses.

While many interventions by Human-Computer Interaction (HCI) researchers and designers have found success in engaging distributed audiences [1, 9, 16], communicating audience response still remains a challenge to translate to online platforms. To help facilitate the expression of audience responses into the digital realm on a live streaming platform like Twitch, we look to biometric sensing devices that capture and convey physiological data. For this demonstration, we will facilitate an interactive experience which allows audience participation remotely through physiological signals (using a tool that facilitates remote heart rate measurement that we call "*Commons Sense*") as a way to heighten audience engagement and effectively capture and communicate the audience experience. We created a top-down survival shooter game that—using webcam-based heart rate detection software—takes an average of each audience member's heart rate as input to affect in-game variables such as lighting and sound design, and game difficulty. The hybrid nature of CHI 2022 will be supported by allowing remote audience attendees to participate as active spectators.

2 BACKGROUND

Physiology has been shown to be an effective tool that has "opened up a new channel of communication between players and spectators" [13]. In the context of digital gaming, physiological interaction (i.e., "affective gaming") has contributed to innovations such as how players provide input to games (e.g., using facial expressions to control movements of an avatar [11]), how games adapt and respond to player emotional state (e.g., adding "horror-like" elements to the game based on players' galvanic skin response (GSR) and electrocardiogram (EKG) data [2]), and how these games might contribute to advances in the mental and physical health domains (e.g., an

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¹<https://www.whatsapp.com/>

²<https://www.twitch.tv/>

³<https://zoom.us/>

electromyography-based dancing game to help people with rehabilitation for ALS patients [18], or a breathing biofeedback game to help children with cystic fibrosis [3]).

Audience participation has long been used in theatre/performance as a way to engage audiences, and is now being adopted into the technological domains. Audience participation games (APGs)—or games that have a “mutually-aware group of audience members who participate in a way that has a meaningful impact on the game” [16]—blurs the lines between streamer and audience. In 2014, an APG called *Twitch Plays Pokémon* gave a global audience control of a game by turning the Twitch live chat into a game controller, allowing audience members to type certain phrases into chat that mapped with game controller inputs in order to advance in the game [12]. Twitch streamers often use home-brew methods to allow audience participation (e.g., chat polls deciding what weapons they should use in-game or what game they should play next), which further demonstrates the value of this mechanic. Despite this, there is still a lack of formal incorporation of audience participation into the Twitch platform [7]. Researchers in the field of Human-Computer Interaction (HCI) have even made tools that enable audience participation including customized emojis for use in the chat [13], interesting heads up display (HUD) elements (e.g., for the spectators to give hints, or vote) [13], expressive visual viewer input [4], and games where the audience plays a role in the game itself [5]. Lessel et al. created an add-on to the video game *Hearthstone* that enables a new communication channel between streamers and their spectators [9], allowing spectators a hands-on way to engage in the gameplay experience, and Fanzo et al. created an audience-driven horror game that requires audience members to use the live chat to help guide the streamer through the game [5]. Off-the-shelf game titles have likewise attempted to incorporate audience participation—both *Tomb Raider* (2013) and *Choice Chamber* (2015) allow spectators to vote on game events, mechanics, and player progression [16].

3 CHI 2022 INTERACTIVE EXPERIENCE

3.1 Game Design and Commons Sense Implementation

There are two parts of the system: a bespoke ‘Alien Shooter’ game and Commons Sense, our novel communication modality. Alien Shooter is a top-down survival shooter set in a dark, outer space-like environment, designed to be streamed on the Twitch platform. The player’s goal is to survive as long as possible, while being attacked by alien enemies (see Figure 1). For each Twitch audience member tuning in, an individual heart rate value is taken using custom software that detects heart rate through a standard webcam, which is then sent to a database. There, the heart rate values for all audience members are averaged and this value is sent back to influence the game input mappings—by manipulating the game *environment* and the game *difficulty* in a form of novel physiological audience communication that we call ‘Commons Sense’.

At CHI 2022, the proposed demonstration will consist of one author playing the Alien Shooter game (either at the live event, or remotely from home), and remote audience attendees. The streamer/player will stream the game to a Twitch channel, with a URL link to a

custom webpage sent out to remote attendees. Attendees will navigate to the custom webpage, which includes an embedded Twitch stream of the streamer through a browser window (cross-browser compatibility is supported). The embedded Twitch stream will allow audience attendees to login through Twitch’s API and continue communication using the Twitch chat. Audience attendees will then be redirected to an instructional page to help with the ideal set-up for the webcam’s heart rate tracking. Audience members will then be asked to provide access to their webcam for the heart rate tracking software. Granting access opened the heart rate tracking in a separate window to allow the process to persist throughout the entire session. This window will notify the audience member of whether the heart rate detection software was currently tracking their HR or not. The demonstration will also allow for attendees at the in-person event to act as the streamer.

3.2 Sensing Audience Heart Rate

We collect Blood Volume Pulse (BVP)—which represents the contractions of the heart in local blood vessels, and thus can be extrapolated as a measure of heart rate (HR)—through attendees’ webcams, as this method is both technologically feasible, and requires minimal setup and calibration from the audience. Heart rate is gathered from each member of the audience watching at home using custom JavaScript software developed by Johanson et al [8]. This software makes use of the headtrackr library [10], which is a library for real-time face tracking and head tracking, tracking the position of a user’s head in relation to the computer screen. Once the head is successfully tracked, the measurement of heart rate is initialized. This is accomplished by detecting small color changes of the forehead, which are mapped to pulse. An event is triggered every time new heart rate data is available. The HR data contains a boolean if the data is valid, the time of the measured HR (in form of javascript `Date.Now()` timestamp), the HR in beats per minute (BPM), the quality of the frequency spectrum (represented as kurtosis), and the data quality as a relation between possible number of datapoints and number of datapoints with a minimum standard deviation as well as the status of the detection of the head [10].

Before using Commons Sense, all spectators are required to undergo a calibration phase of the heart rate detection software, to ensure their webcam is working and all conditions are met for consistent tracking of HR throughout the study (e.g., well lit environment, hair out of face, webcam positioned toward face). Audience members are also provided with a notification on-screen when their webcam is not tracking their HR any longer, giving them opportunity to resolve the issue. Each audience member’s heart rate value is collected (approximately every 2 seconds) and sent to a Firebase database. Every 2 seconds, the heart rate values for all viewers are averaged and then sent to the game. To preserve audience anonymity and privacy, no webcam footage is streamed, recorded, or stored. We also gather a status variable that is used to determine whether or not the audience member should be included when calculating the average. For example, if the forehead is not within view of the webcam and was in the process of re-detection, then the audience member is momentarily be excluded from the average until their heart rate could be retrieved by the system once again. The BPM is added to a total BPM for all audience members

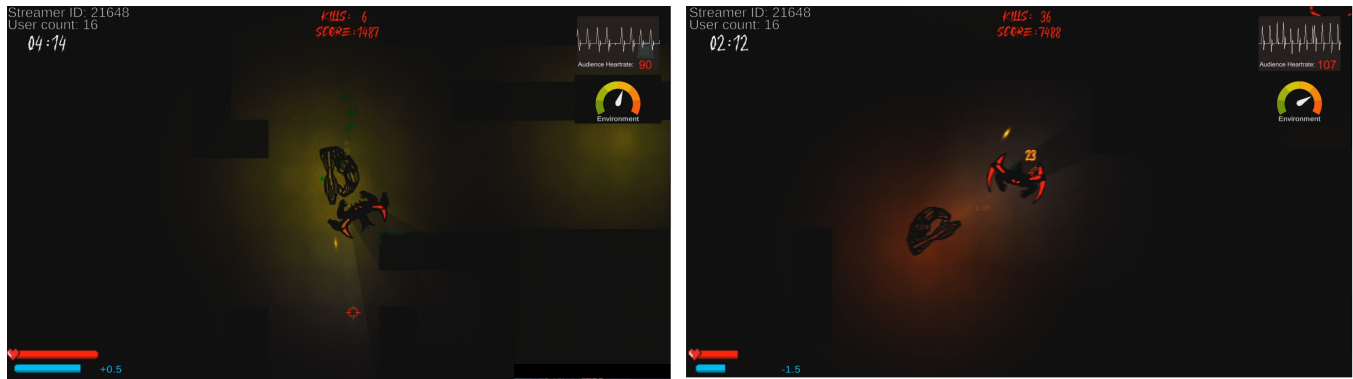


Figure 1: Gameplay footage. Left image depicts the environmental effects with low heart rate and the right image shows environmental effects of a high heart rate.

in the system. A separate back-end system pulls the total BPM and audience member count to calculate the average heart rate among all audience members. Every 500ms the server collects current audience member count as well as the number of audience members, generates an average value by calculating the mean of all the values received, and uploads it to the Firebase database. The average heart rate (HR) is calculated by dividing the sum of all active participant's heart rates by the number of active participants.

3.3 Representing Audience Heart Rate

The audience's average heart rate is both visible and audible throughout the game experience in real-time. The visuals for the heart rate are mapped through the heart rate monitor, difficulty gauge, enemy difficulty, and environment color intensity (see Figure 1). The audio cues are also mapped with the heart rate monitor, allowing the streamer and audience to hear heartbeat audio representing the averaged audience heart rate. The overall effect of visuals and audio demonstrates to both streamers and their audience how much influence the audience's heart rate had on the play experience at any given moment. Through extensive play testing conducted by the research team throughout the implementation process, a minimum heart rate value of 40BPM, and a maximum value of 140BPM were selected to represent the low and high cases of the heart rate percentage—as the mean heart rate would most commonly be within the range of the selected cases. Thus, 40bpm would be mapped to represent the heart rate having 0% effect on the game, and 140BPM would be mapped to represent the heart rate having 100% effect on the game.

3.4 Input Mappings and Influence on Game

Although we represent audience heart rate both visually and sonically, we also aim to use sensed heart rate to directly affect the game. Based on a recent systematic review that explored the field of affective gaming [14], and an in-the-wild study of our game with several popular game streamers [15], we map the audience's heart rate to both environmental and difficulty game effects. The *environmental* effects include those such as lighting intensity and colour. A colour gradient from red to green (see Figure 1, bottom left) was mapped to a low (green) or high (red) average heart rate

value. Green was selected to represent a calming environment and would be mapped to the minimum heart rate effect percentage (0%). Red was selected to represent the horror-inspired environment and would be mapped to the maximum heart rate effect percentage (100%). As the audience's collective heart rate rose, the backlight on the map dimmed and turned red. A lower heart rate was mapped to a greener, more consistent, and brighter light. The lighting colors were selected to correspond to the intensity of the heart rate. Additionally, the opacity of the lights also corresponded with the audience's average heart rate, with the lights brightening as the heart rate decreases and dimming as the heart rate increases. The *difficulty* effects include those such as: speed, damage, and attack range of all enemies, using the present percentage of the heart rate influence, e.g., $\text{current damage} = \text{base damage} + (\text{base damage} * \text{heart rate percentage} / 100)$. The base values for the speed, damage, and attack range were obtained through playtesting and were determined to be the most appropriate variable values for a noticeable difficulty increase. The alien enemies' difficulty variable would be set to their baseline when the heart rate was at 0% and would increment towards the max variable as the mean heart rate increased toward 100%. Thus, when the heart rate percentage was at 0% the damage would be set to the base damage; when the heart rate percentage was at 100% the damage would be set to double the base damage.

3.5 NPC Enemies and Player Health

Choices regarding elements of the game (i.e., design, game balance) were made based on the feedback received from iterative playtesting sessions. Challenge in the game was created through the artificial intelligence (AI) non-player character (NPC) alien enemies. We implemented three enemy types in the game to introduce variety: melee, range, and screamer. The melee and range enemies were designed to have most of the general functionality for a top-down shooter, but also aligned with our core design principals of a simple game design to draw focus onto *Commons Sense* rather than the gameplay. Thus, both melee and range enemies have a simple bi-conditional state manager, with their states corresponding to 'roam' or 'attack'. To make the melee behaviour more engaging, enemies were given a dash mechanic with a cooldown, allowing

their movement speed to increase for a short duration and then return to the default until the cooldown had been reached. The screamer enemy remains stationary and instead applies a buff to all active enemies, multiplying the speed and damage of all range and melee enemies by two.

The health-point (HP) gauge and energy gauge were the main two components affecting the player's in-game character. The HP, initially set to the max of 100, would be subtracted from each time the player took damage, resulting in player death once HP reached 0. Upon death, players would be re-spawned after 4 seconds; re-spawns were designed to be unlimited as to make sure the player kept playing for the entire 5-minute play session. The player's energy—which regenerates over time—was inherently the player's most valuable resource. Energy was used to shoot, activate the flashlight, and sprint. The player was given a gun, thus shooting was the only means of killing enemies. Enabling the flashlight would somewhat brighten the room, display a bright cone of light in front of the player, and make the enemies glow—thus serving as an important tool for visualizing enemies and the environment. While moving around the map remained possible without the use of the flashlight, it became difficult to see enemies hiding in the dark, thus making the chance of being overwhelmed by a group of enemies far more likely. Sprinting would allow the player to escape hazardous situations.

4 CONCLUSION

In summary, we created an audience participation Alien Shooter game which leverages a novel physiological communication modality called 'Commons Sense', as a way of translating lost in-person social cues onto Twitch. Sharing physiological signals as a way to communicate audience response shows promise not only as an engaging and robust form of communication in an impoverished digital domain like Twitch, but also in the broader landscape of HCI. In this hybrid conference, CHI attendees will be able to watch the game and have their heart rate sensed and taken to effect gameplay.

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